SSC MTS 2024 question paper with answers

30 September 2024 · Shift 2

80 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 10 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 16 questions

Q.1

A policeman chases a thief. The speed of the policeman and thief are 10 kmph and 8 kmph, respectively. If the policeman started 15 minutes late, find the time taken by the policeman to catch the thief.

(A) 60 minutes
(B) 15 minutes
(C) 45 minutes
(D) 30 minutes

Q.2

A shot-put made of iron with a diameter of 12 cm is melted into a solid cylinder with a height of 2 cm. What is the diameter of the cylinder made?

(A) 18 cm
(B) 21 cm
(C) 24 cm
(D) 12 cm

Q.3

An oil dealer had three types of oil. 403 litres of the first kind, 434 litres of the second kind and 465 litres of the third kind. Find the maximum volume of oil, filled in each barrel, if equal number of barrels are filled.

(A) 31 litres
(B) 41 litres
(C) 52 litres
(D) 48 litres

Q.4

The marked price of a scooter is ₹1,80,000. The shopkeeper gave a scheme discount of 4% on ₹1,20,000 and 2.5% on the remaining amount. The selling price (in ₹) of the scooter is:

(A) 1,37,700
(B) 1,73,070
(C) 1,73,700
(D) 1,77,300

Q.8

Rohan deposited ₹500 for 2 years, ₹650 for 4 years and ₹1,300 for 6 years, at the same rate of interest. He received the total simple interest of ₹1,140. What is the rate of interest per annum?

(A) 11%
(B) 10.5%
(C) 9.5%
(D) 10%

Q.9

In an election contested by two candidates, one candidate got 25% of the total votes and still lost by 20,000 votes. Find the total number of votes cast? (all votes are valid)

(A) 55000
(B) 35000
(C) 25000
(D) 40000

Q.10

A policeman follows a thief, who is 150 metres ahead of him., and they run at the speed of 6 km/h and 5 km/h, respectively.What distance does the policeman cover to catch up with the thief?

(A) 870 metres
(B) 880 metres
(C) 890 metres
(D) 900 metres

Q.11

A dishonest shopkeeper claims to sell rice at the cost price of ₹95 per kg, but the weight he uses has 1 kg written on it, while it actually weighs 950 gm. The profit he thus earns on selling rice having an actual weight of 95 kg rice is:

(A) ₹375
(B) ₹275
(C) ₹475
(D) ₹500

Q.13

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 6% per year rate of interest for 3 years.

(A) 390
(B) 490
(C) 450
(D) 410

Q.14

A right circular cone of height h and radius R is cut by a plane parallel to the base at a distance h/4 from the base. The ratio of the curved surface areas of the resulting cone to the frustum is:

(A) 9 : 7
(B) 16 : 7
(C) 7 : 16
(D) 16 : 9

Q.15

Find the area of a square whose diagonal is of 14 cm length.

(A) 94 cm^2
(B) 95 cm^2
(C) 98 cm^2
(D) 96 cm^2

Q.16

The mean (rounded off to two decimal place) of the first 6 odd prime numbers is:

(A) 10.71
(B) 8.61
(C) 7.93
(D) 9.33

Q.17

The average of the sales of an electronics shop in the years 1991, 1992, 1993 and 1994 is 2000, and that for the years 1992, 1993, 1994 and 1995 is 2200. If the sales in 1995 are 1800, find the sales in 1991.

(A) 1200
(B) 500
(C) 800
(D) 1000

Q.18

In the first 5 overs of a T-20 international cricket match, the run rate was only 5.6. What should be the run rate (approximately) in the remaining 15 overs to reach a target of 185 runs?

(A) 9.55
(B) 10.25
(C) 10.46
(D) 11.15

Q.19

In a mock test conducted by a coaching institute, 20 students average scored 75% marks and the remaining 15 students average scored 54% marks. Find the average score.

(A) 66%
(B) 62%
(C) 63%
(D) 67%

Q.20

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 5% per year rate of interest for 5 years.

(A) 815
(B) 875
(C) 915
(D) 835

Reasoning Ability and Problem Solving 14 questions

Q.1

TAHO is related to XELS in a certain way based on the English alphabetical order. In the same way, HOVC is related to LSZG. To which of the following is NUBI related, following the same logic?

(A) RYFM
(B) RYKM
(C) RYGM
(D) FYKM

Q.2

12 is related to 59 following a certain logic. Following the same logic, 17 is related to 79. To which ofthe following is 24 related, following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 107
(B) 96
(C) 101
(D) 117

Q.5

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (20, 76, 91) (87, 41, 123)

(A) (40, 80, 15)
(B) (57, 18, 100)
(C) (47, 19, 51)
(D) (84, 47, 126)

Q.7

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 140 × 2 – 4 ÷ 56 + 82 = ?

(A) 268
(B) 241
(C) 262
(D) 212

Q.8

What should come in place of the question mark (?) in the given series based on the English alphabetical order? EGC, IFG, MEK, QDO, ?

(A) WSU
(B) UCS
(C) SWU
(D) VWS

Q.10

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s). Statements: Some metals are iron. Some iron are steel. Conclusions: (I) Some metals are steel. (II) All steel are iron.

(A) Neither conclusion (I) nor (II) follows.
(B) Only conclusion (II) follows.
(C) Both conclusions (I) and (II) follow.
(D) Only conclusion (I) follows.

Q.11

Rajan, Pragyan, Kamal, Sachin, Ashok, Narendra, and Laksh are sitting around a circular table, facing the centre (but not necessarily in the same order). Rajan sits 3rd to the right of Narendra. Kamal sits 2nd to the right of Rajan. Ashok is an immediate neighbour of Narendra. Pragyan sits to the immediate right of Sachin. Who is sitting 3rd to the left of Narendra?

(A) Sachin
(B) Laksh
(C) Pragyan
(D) Kamal

Q.12

In a certain code language, 'FAST' is coded as '6842' and 'TILE' is coded as '5761'.What is the code for 'T' in the given code language?

(A) 5
(B) 1
(C) 6
(D) 2

Q.13

In a certain code language, 'checked his mirror' is written as 'aw ar ae' and 'book inevitably mirror' is coded as 'al aw ao'. How is 'mirror' coded in that language?

(A) al
(B) aw
(C) ae
(D) ao

Q.14

P, Q, R, S, T, U, V and W are sitting around a circular table facing the centre(but not necessarily in the same order). P is sitting to the immediate right of V and third to the left of U. S is sitting fourth to the right of Q, who is not an immediate neighbour of P and T. Q is sitting to the second right of U. R is sitting to the immediate right of Q and second to the left of P. How many people are sitting between R and T when counted from the left of T?

(A) Four
(B) Five
(C) Three
(D) Two

Q.15

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some hooks are rags. All rags are bottles. Conclusions: (I) Some bottles are hooks. (II) All bottles are rags.

(A) Only conclusion (II) follows
(B) Both conclusions (I) and (II) follow
(C) Neither conclusion (I) nor (II) follows
(D) Only conclusion (I) follows

Q.17

What should come in place of the question mark (?) in the given series? 34 51 85 102 136 ?

(A) 153
(B) 157
(C) 142
(D) 146

Q.18

GMIE is related to HNJF in a certain way based on the English alphabetical order. In the same way, KQMI is related to LRNJ. To which of the following is NTPL related, following the same logic?

(A) OUMQ
(B) UOMQ
(C) UOQM
(D) OUQM

Q.20

In a certain code language, ‘will make you’ is written as ‘ft fy fd’ and ‘you see what’ is writtenas ‘fi fo ft’. How will ‘you’ be writtenin the given language?

(A) fd
(B) ft
(C) fo
(D) fy

General Awareness 25 questions

Q.1

Calculate the revenue receipts from the following information. Revenue deficit= 2.80% of GDP Revenue expenditure= 10.80% of GDP

(A) 13.60
(B) 12.80
(C) 8.00
(D) 2.70

Q.2

How many cities have been selected to be developed as sustainable and responsible destinations as per the Swadesh Darshan 2.0 scheme launched by the Union Ministry of Tourism in January 2023?

(A) 21
(B) 67
(C) 30
(D) 45

Q.3

In 1928, who among the following wrote the Congress Party’s Nehru Report, a future constitution for independent India, based on the granting of dominion status?

(A) Subhash Chandra Bose
(B) Mahatma Gandhi
(C) Jawaharlal Nehru
(D) Motilal Nehru

Q.4

The autobiography 'Wings of Fire' chronicles the life of which notable Indian personality?

(A) Kiran Bedi
(B) Arundhati Roy
(C) Ratan Tata
(D) APJ Abdul Kalam

Q.5

In animal cells, the breakdown of very long chain fatty acids through β-oxidation is conducted by which of the following organelles?

(A) Peroxisome
(B) Golgi apparatus
(C) Mitochondria
(D) Ribosome

Q.6

Which of the following correctly describes the nature of the Indian Constitution?

(A) Rigid
(B) Flexible
(C) Very rigid and difficult to amend
(D) Neither rigid nor flexible

Q.7

Which programme, that was introduced in the year 2000-2001, aims to offer homes, drinking water, primary education and basic healthcare in rural areas?

(A) Pradhan Mantri Kisan Samman Nidhi
(B) Pradhan Mantri Vaya Vandana Yojana
(C) Pradhan Mantri Gramodaya Yojana
(D) The Pradhan Mantri Gram Sadak Yojana

Q.8
Which of the following dances is traditionally performed by the males of three adjacent states - Jharkhand, Odisha and West Bengal?

(A) Kuchipudi
(B) Odissi
(C) Chhau
(D) Sattriya

Q.9
Which of the following revolutionaries, who was arrested in Lahore conspiracy case by the British Government of India, was martyred in the hunger strike of 63 days in jail?

(A) Jatindra Nath Das
(B) Ram Prasad Bismil
(C) Rash Behari Bose
(D) Prafulla Chaki

Q.10
The term 'foot fault' is associated with which of the following games?

(A) Basketball
(B) Hockey
(C) Tennis
(D) Archery

Q.11
Which of the following dams is a part of the Damodar Valley Corporation?

(A) Rihand
(B) Koyna
(C) Salal
(D) Maithon

Q.12
When a large amount of carbon dioxide dissolves in a water body, the water becomes acidic because of the formation of which of the following acids?

(A) Carbonic acid
(B) Carbolic acid
(C) Carboxylic acid
(D) Carbamic acid

Q.13
Select the INCORRECT pair of molecule/formula units and their molar mass from the following.

(A) NaCl - 58.5
(B) CH_4 - 16
(C) HCl - 36.5
(D) HNO_3 - 62

Q.14
The Patkai hills are part of the _____ mountains.

(A) Purvachal
(B) Eastern Ghats
(C) Western Ghats
(D) Karakoram

Q.15
In which of the following events did India get its first individual Olympic gold medal?

(A) Boxing
(B) Weightlifting
(C) Wrestling
(D) Shooting

Q.16
In the 1991 Census, which state was the highest in rank in terms of literacy rate and held higher than the national average of literacy rate?

(A) Tamil Nadu
(B) Karnataka
(C) West Bengal
(D) Kerala

Q.17
Raga Deepak has been associated with which famous musician?

(A) Lal Kalawant
(B) Ali Khan Karori
(C) Tansen
(D) Khusrau

Q.18
As per Article 181, the _____ can not preside while a resolution for his removal from office is under consideration in State Legislative Assembly.

(A) Advocate General
(B) Speaker
(C) Deputy Chairman
(D) Chairman

Q.19
The 'Bathukamma' festival is associated with which state?

(A) Sikkim
(B) Punjab
(C) Tripura
(D) Telangana

Q.20
Which organelle contains ribosomal RNA and is involved in the production of ribosomes?

(A) Endoplasmic reticulum
(B) Nucleus
(C) Nucleolus
(D) Golgi apparatus

Q.21
Who was the court poet of king Harshavardhana?

(A) Vatsbhatti
(B) Ravikirti
(C) Kalidas
(D) Banabhatta

Q.22
Balban’s rule as an absolute despot led to which of the following events?

(A) Granting Jagirs to Nobles
(B) Respected common man
(C) Destruction of the Forty
(D) Appointed Hindus in administration

Q.23
Which of the following institutes/projects is declared as a masterpiece of oral and intangible heritage of humanity by UNESCO in May 2001?

(A) National Project to support Kuttiyattam
(B) Music Academy Madras
(C) Jawaharlal Nehru Manipur Dance Academy
(D) National Institute of Kathak Dance

Q.24
In September 2023, after the retirement of Incumbent Sanjay Kumar Mishra, who was appointed as In-charge Director of ED?

(A) TV Somanathan
(B) Tushar Mehta
(C) Rahul Navin
(D) Samant Goel

Q.25
Which of the following architectural elements is excluded from the construction sites of the Indus Valley Civilisation?

(A) Baked Bricks
(B) Mortar of Gypsum
(C) Iron Slabs
(D) Sun Dried Bricks

English Language and Comprehension 25 questions

Q.1

One of the four underlined words in the given sentence is incorrectly spelt. Select the INCORRECTLY spelt word. Let us “appreciate”, support and “contribute” our might to the “promotional” efforts of the “comunity”.

(A) appreciate
(B) contribute
(C) comunity
(D) promotional

Q.2

Select the option that has an error in the use of phrasal verb.

(A) Clean up the mess in the kitchen. It is unhygienic.
(B) We look forward to seeing you during the event.
(C) To earn a significant profit, you are advised to deal in electronic items.
(D) Why does she break out with all her near and dear ones?

Q.3

Select the most appropriate ANTONYM of the underlined word in the given sentence. The company plans to “expand” its operations into global markets soon.

(A) Spread
(B) Diversify
(C) Restrict
(D) Eliminate

Q.4

Select the most appropriate ANTONYM of the given word. Triumph

(A) Lapse
(B) Delay
(C) Failure
(D) Attentiveness

Q.5

Select the most appropriate meaning of the underlined word. I cannot see without my “glasses”.

(A) An unbreakable object
(B) A device used to correct the defects of vision
(C) A fragile object
(D) A container

Q.6

Select the most appropriate meaning of the idiom from the options given below. To show the white feather

(A) To act cowardly
(B) To maintain peace
(C) To act bravely
(D) To show anger

Q.7

Select the correct spelling of the underlined word in the given sentence. After their disagreement, they decided to “saparate” their belongings and part ways.

(A) seperate
(B) separate
(C) seprete
(D) seprate

Q.8

Select the grammatically correct sentence from the given options.

(A) An unexpected power failure completely foiled an plan of the team.
(B) An unexpected power failure completely foiled the plan of the team.
(C) A unexpected power failure completely foiled the plan of a team.
(D) An unexpected power failure completely foiled a plan of a team.

Q.9

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. My brother trains young girls at the football academy, whereas I gets a handsome salary to teach mathematics at the same place.

(A) to teach mathematics at the same place
(B) I gets a handsome salary
(C) at the football academy, whereas
(D) My brother trains young girls

Q.10

In the following sentence the underlined part contains as error. Alternatives to the underlined part are given as options. Select the correct alternative. The bread was “drier” and stale.

(A) much drier
(B) driest
(C) more drier
(D) dry

Q.11

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in the blanks. Johannes Gutenberg invented the first printing press that could print thousands of _____ a day. This made books cheaper and information _____ widely available.

(A) sheet; much
(B) sides; most
(C) leaf; many
(D) pages; more

Q.12

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. All in all, he had the “more splendid” collection of antique paintings ever.

(A) most splendid
(B) splendid
(C) splendidest
(D) more splendider

Q.13

Select the most appropriate option to fill in the blank. The scientist conducted a series of experiments to _____ his hypothesis.

(A) variable
(B) vague
(C) verify
(D) variation

Q.14

Select the most appropriate synonym of the given word. Reverent

(A) Rushed
(B) Important
(C) Respectful
(D) Related

Q.15

Select the most appropriate option that can substitute the underlined segment in the given sentence. I “long from travel” outside the country, dreaming of adventures among the towering Alps and basking in the turquoise waters of the Caribbean.

(A) long to travel
(B) long for travelling
(C) long for travelled
(D) long at travel

Q.16

Rectify the sentence by selecting the correct spelling from the options. All the political parties of India appaer lost in the physical exercise of fighting elections.

(A) Appear
(B) Apeear
(C) Aapear
(D) Apearr

Q.17

Select the option that can be used as a one-word substitute for the given group of words. A building where animals are butchered

(A) Burrow
(B) Cemetery
(C) Decanter
(D) Abattoir

Q.18

Select the most appropriate ANTONYM that can substitute the underlined segment in the given sentence. They all felt that the movie was truly “captivating” and unforgettable.

(A) charismatic
(B) repugnant
(C) relatable
(D) tantalising

Q.19

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Although she is usually / rude with everyone, / she behaved nice / with all of us today.

(A) rude with everyone
(B) with all of us today
(C) she behaved nice
(D) Although she is usually

Q.20

Select the most appropriate idiom that can substitute the underlined segment in the given sentence. She is “someone who always discourages our enthusiasm;” therefore, we have decided to keep our distance from her.

(A) a wet blanket
(B) a nine days' wonder
(C) a cash cow
(D) a fish out of water

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Goddess Shailputri is the first form of Goddess Durga, who is worshipped ____(1)___ Navaratri. In this form, she is worshipped as the daughter of ___(2)__ Himalayas. She is the goddess ___(3)____ nature and is adorned with flowers. She ___(4)__ a bull and holds a trident in one hand and a lotus flower in another. During Navaratri, nine different forms of Durga are worshipped. Durga is a ___(5)__, ___(6)___, ___(7)___ deity who is also known as Adishakti—the energy that manifested to create the universe.

Select the most appropriate option to fill in blank number 1.

(A) during
(B) among
(C) to
(D) by

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Goddess Shailputri is the first form of Goddess Durga, who is worshipped ____(1)___ Navaratri. In this form, she is worshipped as the daughter of ___(2)__ Himalayas. She is the goddess ___(3)____ nature and is adorned with flowers. She ___(4)__ a bull and holds a trident in one hand and a lotus flower in another. During Navaratri, nine different forms of Durga are worshipped. Durga is a ___(5)__, ___(6)___, ___(7)___ deity who is also known as Adishakti—the energy that manifested to create the universe.

Select the most appropriate article to fill in blank number 2.

(A) an
(B) a
(C) no article
(D) the

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Goddess Shailputri is the first form of Goddess Durga, who is worshipped ____(1)___ Navaratri. In this form, she is worshipped as the daughter of ___(2)__ Himalayas. She is the goddess ___(3)____ nature and is adorned with flowers. She ___(4)__ a bull and holds a trident in one hand and a lotus flower in another. During Navaratri, nine different forms of Durga are worshipped. Durga is a ___(5)__, ___(6)___, ___(7)___ deity who is also known as Adishakti—the energy that manifested to create the universe.

Select the most appropriate option to fill in blank number 3.

(A) around
(B) at
(C) with
(D) of

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Goddess Shailputri is the first form of Goddess Durga, who is worshipped ____(1)___ Navaratri. In this form, she is worshipped as the daughter of ___(2)__ Himalayas. She is the goddess ___(3)____ nature and is adorned with flowers. She ___(4)__ a bull and holds a trident in one hand and a lotus flower in another. During Navaratri, nine different forms of Durga are worshipped. Durga is a ___(5)__, ___(6)___, ___(7)___ deity who is also known as Adishakti—the energy that manifested to create the universe.

Select the most appropriate option to fill in blank number 4.

(A) was riding
(B) had ridden
(C) rode
(D) rides

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Goddess Shailputri is the first form of Goddess Durga, who is worshipped ____(1)___ Navaratri. In this form, she is worshipped as the daughter of ___(2)__ Himalayas. She is the goddess ___(3)____ nature and is adorned with flowers. She ___(4)__ a bull and holds a trident in one hand and a lotus flower in another. During Navaratri, nine different forms of Durga are worshipped. Durga is a ___(5)__, ___(6)___, ___(7)___ deity who is also known as Adishakti—the energy that manifested to create the universe.

Select the most appropriate option to fill in blanks number 5.

(A) Indian; radiant; powerful
(B) radiant; Indian; powerful
(C) powerful; radiant; Indian
(D) powerful; Indian; radiant

Answer key

Numerical and Mathematical Ability

Q.1 **(A)**	Q.2 **(C)**	Q.3 **(A)**	Q.4 **(C)**	Q.8 **(D)**	Q.9 **(D)**
Q.10 **(D)**	Q.11 **(C)**	Q.13 **(C)**	Q.14 **(A)**	Q.15 **(C)**	Q.16 **(D)**
Q.17 **(D)**	Q.18 **(C)**	Q.19 **(A)**	Q.20 **(B)**		

Reasoning Ability and Problem Solving

Q.1 **(A)**	Q.2 **(A)**	Q.5 **(D)**	Q.7 **(D)**	Q.8 **(B)**	Q.10 **(A)**
Q.11 **(B)**	Q.12 **(C)**	Q.13 **(B)**	Q.14 **(C)**	Q.15 **(D)**	Q.17 **(A)**
Q.18 **(D)**	Q.20 **(B)**				

General Awareness

Q.1 **(C)**	Q.2 **(C)**	Q.3 **(D)**	Q.4 **(D)**	Q.5 **(A)**	Q.6 **(D)**
Q.7 **(C)**	Q.8 **(C)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(D)**	Q.12 **(A)**
Q.13 **(D)**	Q.14 **(A)**	Q.15 **(D)**	Q.16 **(D)**	Q.17 **(C)**	Q.18 **(B)**
Q.19 **(D)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(C)**	Q.23 **(A)**	Q.24 **(C)**
Q.25 **(C)**					

English Language and Comprehension

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(C)**	Q.4 **(C)**	Q.5 **(B)**	Q.6 **(A)**
Q.7 **(B)**	Q.8 **(B)**	Q.9 **(B)**	Q.10 **(D)**	Q.11 **(D)**	Q.12 **(A)**
Q.13 **(C)**	Q.14 **(C)**	Q.15 **(A)**	Q.16 **(A)**	Q.17 **(D)**	Q.18 **(B)**
Q.19 **(C)**	Q.20 **(A)**	Q.21 **(A)**	Q.22 **(D)**	Q.23 **(D)**	Q.24 **(D)**
Q.25 **(C)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (A) 60 minutes

In the 15 minutes the thief gets 8 × 15/60 = 2 km ahead. The policeman gains 10 − 8 = 2 km/h on him, so he takes 2/2 = 1 hour: 60 minutes.

Q.2 · Answer (C) 24 cm

The shot-put's volume is (4/3)π × 6^3 = 288π cm^3. With height 2 cm, πr^2 × 2 = 288π, so r^2 = 144, r = 12 cm and the diameter is 24 cm.

Q.3 · Answer (A) 31 litres

The largest volume that divides all three exactly is their HCF: 403 = 31 × 13, 434 = 31 × 14 and 465 = 31 × 15, so 31 litres.

Q.4 · Answer (C) 1,73,700

Discount = 4% of ₹1,20,000 + 2.5% of ₹60,000 = 4,800 + 1,500 = ₹6,300. Selling price = 1,80,000 − 6,300 = ₹1,73,700.

Q.8 · Answer (D) 10%

(500 × 2 + 650 × 4 + 1,300 × 6) × R/100 = 11,400 × R/100 = 1,140, so R = 10%.

Q.9 · Answer (D) 40000

The winner got 75% and the loser 25%, so the margin, 50% of the votes, is 20,000: 40,000 votes were cast.

Q.10 · Answer (D) 900 metres

The policeman gains 6 − 5 = 1 km/h, so he closes the 150 m (0.15 km) in 0.15 hours. In that time he runs 6 × 0.15 = 0.9 km = 900 metres.

Q.11 · Answer (C) ₹475

Each '1 kg' he sells is 950 g, so 95 kg of rice goes out as 95/0.95 = 100 'kg' for 100 × 95 = ₹9,500. It cost him 95 × 95 = ₹9,025: a profit of ₹475.

Q.13 · Answer (C) 450

SI = 2,500 × 6 × 3/100 = ₹450.

Q.14 · Answer (A) 9 : 7

The cone cut off at the top has height 3h/4, so its radius and slant height are 3/4 of the original's and its curved surface is $(3/4)^2$ = 9/16 of the whole. The frustum has the other 7/16: 9 : 7.

Q.15 · Answer (C) 98 cm^2

Area of a square = $diagonal^2/2$ = $14^2/2$ = 98 cm^2.

Q.16 · Answer (D) 9.33

The first six odd primes are 3, 5, 7, 11, 13 and 17. Their sum is 56, and 56/6 = 9.33.

Q.17 · Answer (D) 1000

The 1991-94 total is 4 × 2000 = 8000 and the 1992-95 total 4 × 2200 = 8800. The difference, 800, is the 1995 sales minus the 1991 sales, so 1991 = 1800 − 800 = 1000.

Q.18 · Answer (C) 10.46

5 overs at 5.6 an over gave 28 runs, so 185 − 28 = 157 are needed from 15 overs: 157/15 = 10.466… an over, and 10.46 is the nearest option.

Q.19 · Answer (A) 66%

(20 × 75 + 15 × 54)/35 = (1,500 + 810)/35 = 2,310/35 = 66%.

Q.20 · Answer (B) 875

SI = 3,500 × 5 × 5/100 = ₹875.

Reasoning Ability and Problem Solving

Q.1 · Answer (A) RYFM

Each letter moves 4 places forward: T → X, A → E, H → L, O → S and H → L, O → S, V → Z, C → G. So N → R, U → Y, B → F and I → M: RYFM.

Q.2 · Answer (A) 107

The second number is four times the first plus 11: 12 × 4 + 11 = 59 and 17 × 4 + 11 = 79. Likewise 24 × 4 + 11 = 107.

Q.5 · Answer (D) (84, 47, 126)

The third number is the sum of the first two minus 5: 20 + 76 – 5 = 91 and 87 + 41 – 5 = 123. Likewise 84 + 47 – 5 = 126.

Q.7 · Answer (D) 212

After interchanging the signs: 140 ÷ 2 + 4 × 56 – 82 = 70 + 224 – 82 = 212.

Q.8 · Answer (B) UCS

First letters move 4 forward (E, I, M, Q, so U); second letters 1 back (G, F, E, D, so C); third letters 4 forward (C, G, K, O, so S): UCS.

Q.10 · Answer (A) Neither conclusion (I) nor (II) follows.

The iron that is metal need not be the iron that is steel, so (I) does not follow; and 'some iron are steel' leaves room for steel that is not iron, so (II) does not follow either.

Q.11 · Answer (B) Laksh

Going round to the right from Narendra: Narendra, Sachin, Pragyan, Rajan, Laksh, Kamal, Ashok. (Rajan is third to his right and Kamal second to Rajan's right; only with Ashok on Narendra's left do Sachin and Pragyan find two adjacent free seats.) Third to Narendra's left is Laksh.

Q.12 · Answer (C) 6

T is the only letter in both words and 6 the only digit in both codes, so T = 6.

Q.13 · Answer (B) aw

'mirror' is the only word in both sentences and 'aw' the only code in both, so mirror = aw.

Q.14 · Answer (C) Three

Going round to the right from P: P, S, T, U, W, Q, R, V. (V is on P's left and U third to P's right; Q is second to U's right, R next to Q and S fourth from Q; T cannot sit beside Q, so T takes the seat between S and U.) Counting leftwards from T, S, P and V come before R: three.

Q.15 · Answer (D) Only conclusion (I) follows

Some hooks are rags and all rags are bottles, so some hooks are bottles: (I) follows. There may be bottles that are not rags, so (II) does not.

Q.17 · Answer (A) 153

The differences alternate 17 and 34 (51 – 34 = 17, 85 – 51 = 34, 102 – 85 = 17, 136 – 102 = 34), so the next is 136 + 17 = 153.

Q.18 · Answer (D) OUQM

Each letter moves 1 place forward: G → H, M → N, I → J, E → F and K → L, Q → R, M → N, I → J. So NTPL → OUQM.

Q.20 · Answer (B) ft

'you' is the only word in both sentences and 'ft' the only code in both, so you = ft.

General Awareness

Q.1 · Answer (C) 8.00

Revenue receipts = revenue expenditure – revenue deficit = 10.80% – 2.80% = 8.00% of GDP.

Q.2 · Answer (C) 30
SSC's key gives 30, the number of cities (in 15 states) reported as shortlisted for the first phase of Swadesh Darshan 2.0. Later official lists of its destinations are longer.

Q.3 · Answer (D) Motilal Nehru
The Nehru Report (1928), which asked for dominion status, was drafted by an All Parties Conference committee chaired by Motilal Nehru, after whom it is named; Jawaharlal Nehru was its secretary.

Q.4 · Answer (D) APJ Abdul Kalam
'Wings of Fire', written with Arun Tiwari, is A.P.J. Abdul Kalam's autobiography.

Q.5 · Answer (A) Peroxisome
Peroxisomes shorten very long chain fatty acids by β-oxidation; mitochondria oxidise the shorter chains.

Q.6 · Answer (D) Neither rigid nor flexible
The Constitution is a blend of rigidity and flexibility: some provisions can be amended by a simple majority of Parliament, others need a special majority, and some also need ratification by half the states.

Q.7 · Answer (C) Pradhan Mantri Gramodaya Yojana
The Pradhan Mantri Gramodaya Yojana, launched in 2000-01, funded primary health, primary education, rural shelter, rural drinking water, nutrition and rural electrification.

Q.8 · Answer (C) Chhau
Chhau is danced by men in three styles: Purulia in West Bengal, Seraikella in Jharkhand and Mayurbhanj in Odisha.

Q.9 · Answer (A) Jatindra Nath Das
Jatindra Nath Das, arrested in the Lahore Conspiracy Case, died in Lahore jail on 13 September 1929, on the 63rd day of his hunger strike.

Q.10 · Answer (C) Tennis
In tennis a foot fault is called when the server's foot touches the baseline or the court inside it before the ball is struck.

Q.11 · Answer (D) Maithon
Maithon Dam, on the Barakar, is one of the Damodar Valley Corporation's dams. Rihand is in Uttar Pradesh, Koyna in Maharashtra and Salal, on the Chenab, in Jammu and Kashmir.

Q.12 · Answer (A) Carbonic acid
Carbon dioxide reacts with water to form carbonic acid (H_2CO_3), which makes the water acidic.

Q.13 · Answer (D) HNO_3 - 62
HNO_3 has a molar mass of $1 + 14 + 3 \times 16 = 63$, not 62. NaCl ($23 + 35.5 = 58.5$), CH_4 ($12 + 4 = 16$) and HCl ($1 + 35.5 = 36.5$) are right.

Q.14 · Answer (A) Purvachal
The Patkai hills, on the India-Myanmar border, belong to the Purvachal, the eastern hills of the Himalayas, with the Naga, Manipur and Mizo hills.

Q.15 · Answer (D) Shooting
Abhinav Bindra won the 10 m air rifle gold at Beijing 2008, India's first individual Olympic gold.

Q.16 · Answer (D) Kerala
In the 1991 Census Kerala had the highest literacy rate, 89.81%, against the national 52.21%.

Q.17 · Answer (C) Tansen
Legend has it that Tansen, the musician of Akbar's court, sang Raga Deepak so well that the lamps lit themselves.

Q.18 · Answer (B) Speaker
Under Article 181 the Speaker (or the Deputy Speaker) of a State Legislative Assembly may not preside while a resolution for his removal from office is under consideration.

Q.19 · Answer (D) Telangana
Bathukamma, a floral festival celebrated by women during Navaratri, is the state festival of Telangana.

Q.20 · Answer (C) Nucleolus
The nucleolus, inside the nucleus, makes ribosomal RNA and assembles the subunits of ribosomes.

Q.21 · Answer (D) Banabhatta
Banabhatta, author of the 'Harshacharita' and 'Kadambari', was Harshavardhana's court poet. Ravikirti served the Chalukya king Pulakeshin II.

Q.22 · Answer (C) Destruction of the Forty
Balban, who ruled as an absolute monarch, broke the power of the Chahalgani, the Forty Turkish nobles.

Q.23 · Answer (A) National Project to support Kuttiyattam
Kutiyattam, the Sanskrit theatre of Kerala, was among the first Masterpieces of the Oral and Intangible Heritage of Humanity proclaimed by UNESCO in May 2001.

Q.24 · Answer (C) Rahul Navin
Rahul Navin, a 1993-batch IRS officer and Special Director of the ED, was made its in-charge Director in September 2023 when Sanjay Kumar Mishra's term ended.

Q.25 · Answer (C) Iron Slabs
The Harappans built with baked and sun-dried bricks and used gypsum mortar, but iron was unknown to them: theirs was a Bronze Age civilisation.

English Language and Comprehension

Q.1 · Answer (C) comunity
The correct spelling is 'community'.

Q.2 · Answer (D) Why does she break out with all her near and dear ones?
To end a relationship with someone is to 'break up with' them; 'break out' means to begin suddenly or to escape.

Q.3 · Answer (C) Restrict
To 'expand' is to make larger; its opposite is 'restrict', to keep within limits.

Q.4 · Answer (C) Failure
The opposite of 'triumph' (success) is 'failure'.

Q.5 · Answer (B) A device used to correct the defects of vision
'Glasses' here are spectacles: lenses in a frame that correct defects of vision.

Q.6 · Answer (A) To act cowardly
To 'show the white feather' is to show cowardice.

Q.7 · Answer (B) separate
The correct spelling is 'separate'.

Q.8 · Answer (B) An unexpected power failure completely foiled the plan of the team.
SSC's key picks 'foiled the plan of the team'. 'An' goes before a vowel sound and 'a' before a consonant sound, so 'an plan' and 'a unexpected' are wrong; 'a plan of a team' is not ungrammatical, but a particular plan of a particular team takes 'the'.

Q.9 · Answer (B) I gets a handsome salary
The subject 'I' takes 'get', not 'gets': 'whereas I get a handsome salary'.

Q.10 · Answer (D) dry
Nothing is being compared, so the positive degree is needed, like 'stale': 'The bread was dry and stale.'

Q.11 · Answer (D) pages; more
'Thousands of' needs a plural noun, 'pages', and the comparison with the time before the press calls for 'more widely available'.

Q.12 · Answer (A) most splendid
'Ever' compares the collection with all others, which needs the superlative: 'the most splendid collection of antique paintings ever'.

Q.13 · Answer (C) verify
Experiments are carried out to 'verify', that is confirm, a hypothesis.

Q.14 · Answer (C) Respectful
'Reverent' means feeling or showing deep respect: 'respectful'.

Q.15 · Answer (A) long to travel
For an action one yearns to do, 'long' takes a to-infinitive: 'I long to travel outside the country'. 'Long for' normally takes a plain noun ('long for home'), and 'long from' and 'long at' are not used.

Q.16 · Answer (A) Appear
The correct spelling is 'appear'.

Q.17 · Answer (D) Abattoir
An abattoir is a slaughterhouse, where animals are killed for meat.

Q.18 · Answer (B) repugnant
'Captivating' means charming and attractive; its opposite is 'repugnant', causing strong dislike.

Q.19 · Answer (C) she behaved nice
The verb 'behaved' needs an adverb: 'she behaved nicely'.

Q.20 · Answer (A) a wet blanket
A 'wet blanket' is someone who spoils other people's enthusiasm or fun.

Q.21 · Answer (A) during
'During' names the period in which something happens: 'worshipped during Navaratri'.

Q.22 · Answer (D) the
Names of mountain ranges take 'the': 'the daughter of the Himalayas'.

Q.23 · Answer (D) of
'Of' links the goddess to what she presides over: 'the goddess of nature'.

Q.24 · Answer (D) rides
The passage describes her in the simple present ('holds a trident'), so 'She rides a bull'.

Q.25 · Answer (C) powerful; radiant; Indian
An adjective of origin follows adjectives of opinion or quality, so 'Indian' comes last: 'a powerful, radiant, Indian deity'.

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